
Chapter 1

Overview on Machine Learning

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Outline

- ❑ Pattern Recognition
- ❑ Machine learning
- ❑ Related fields of pattern recognition
- ❑ Classification

1. Pattern recognition

- Humans are good at recognizing objects (or ***patterns***).
- We find it difficult to write a computer program to recognize objects.
- Ex: By analyzing sample images of faces, a program should be able to capture the pattern specific to a face and identify it as a face. This is ***pattern recognition***.
 - There may be several classes and we have to classify a particular face into a certain category (or *class*). This is ***classification***.
- In pattern recognition, the term *pattern* is used to include all objects that we want to classify.
- A ***class*** is a collection of objects that are similar, but not necessarily identical, and which is distinguishable from other classes.

Figure 1.1 illustrates the difference between classification where the classes are known beforehand and classification where classes are created after inspecting the objects.

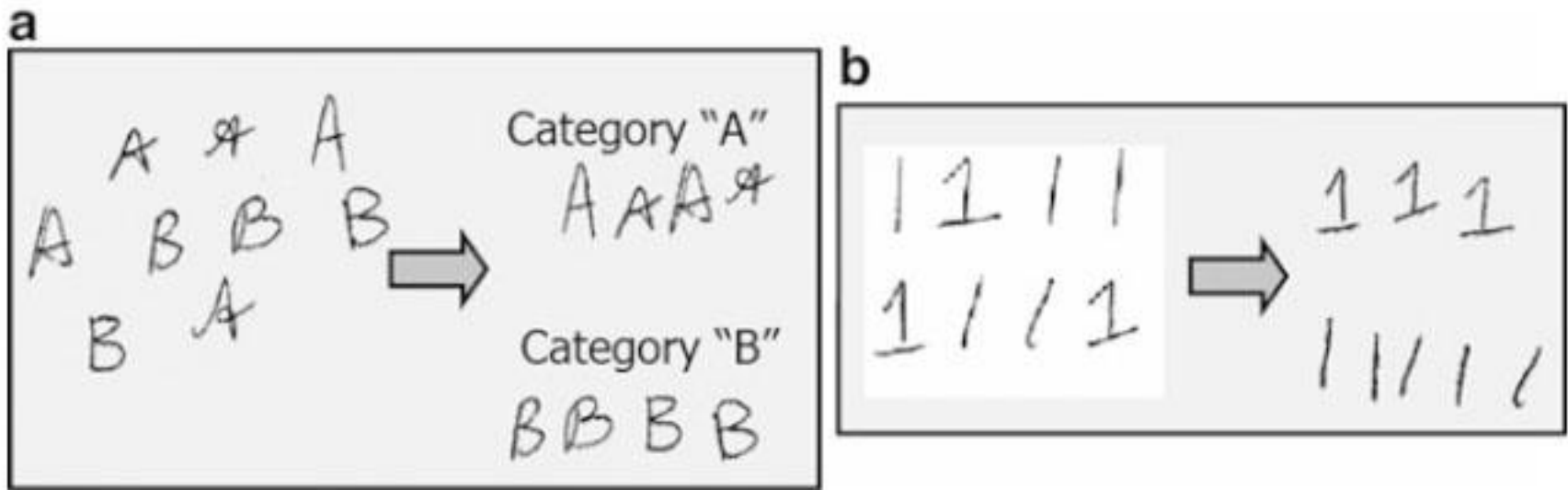


Figure 1.1 Classification when the classes are (a) known and b) unknown beforehand.

Applications of pattern recognitions

Interest in pattern recognition has grown due to emerging applications. These include:

- Data mining
- Biometrics
 - personal identification based on physical attributes of the face, iris, fingerprints, etc.
- Machine vision
 - automatic visual inspection in an assembly line
- Character recognition
 - automatic mail sorting by zip code, automatic check scanners at ATMs.
- Document recognition
 - recognize whether an email is spam or not, based on the message header and content.
- Speech recognition
 - helping handicapped patients to control machines.

- Computer-added diagnosis
 - helping doctors make diagnostic decisions based on interpreting medical data such as ultrasound images, electrocardiograms (ECGs) or electroencephalograms (EEGs).
- Medical imaging
 - classifying cells as malignant or benign based on magnetic resonance imaging (MRI) scans, or classify different emotional and cognitive states from the images of brain activity in functional MRI
- Bioinformatics
 - DNA sequence analysis to detect genes related to particular diseases.
- Remote sensing
 - land use and crop yield
- Astronomy
 - classifying galaxies based on their shapes

2. Machine learning

- What ***learning*** is?
- **Learning from experience**
 - *Remembering, adapting and generalization*
 - *Reasoning and logical deduction*
- ***Machine learning*** is about making computers modify or adapt their actions so that their actions get more accurate, where accuracy is measured by how well the chosen actions reflect the correct ones.
- Through *learning*, computers can *recognize* patterns correctly.
- So *machine learning* “helps” *pattern recognition*. Without machine learning, computers can not achieve pattern recognition.

Some paradigms of machine learning

■ Supervised learning

- A training set of *examples* (patterns) with the correct *responses* (class labels) are provided and based on this training set, the algorithm *generalizes* to respond correctly to all possible inputs. This is also called learning from ***exemplars***.

■ Unsupervised learning

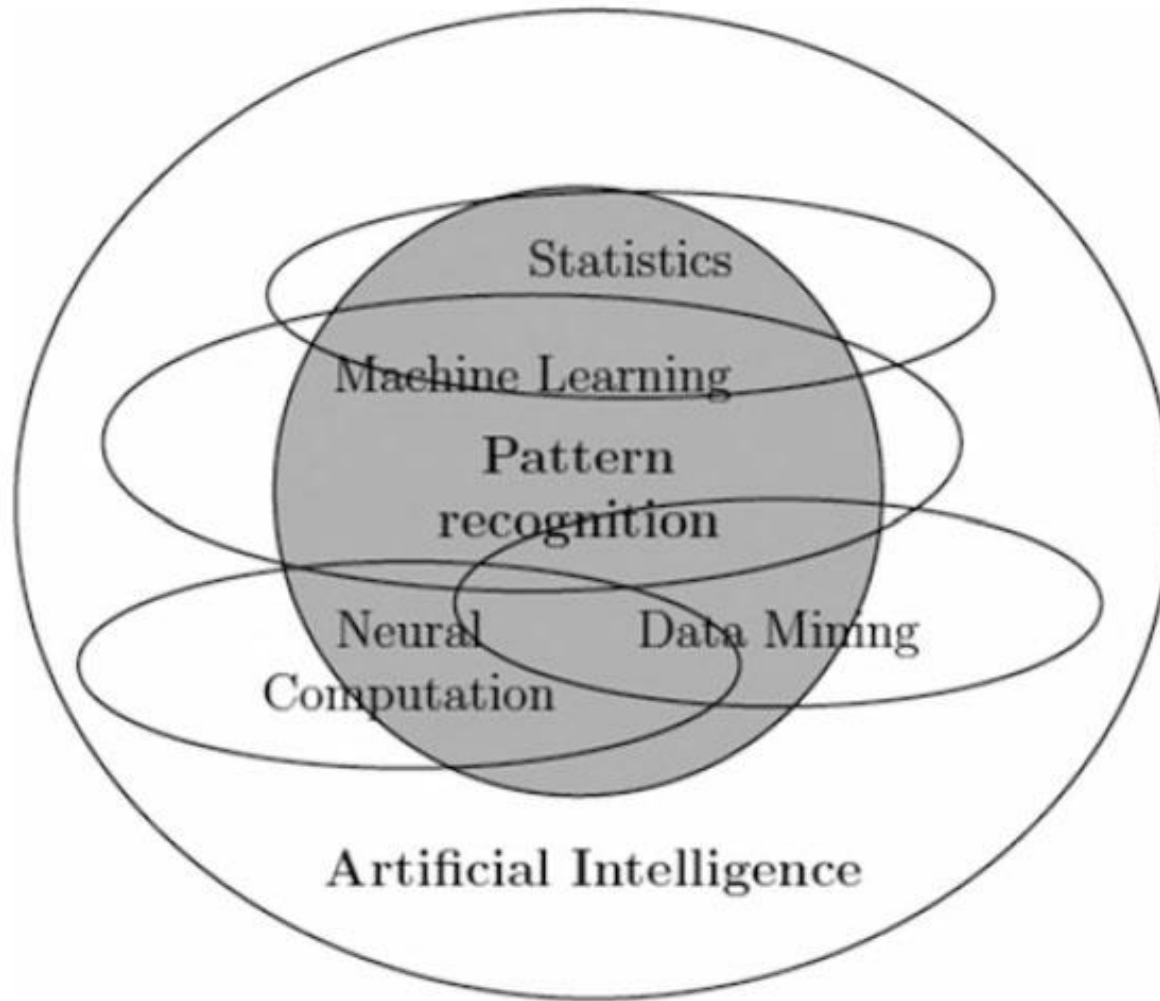
- Correct responses (class labels) are not provided, instead the algorithm tries to identify similarities between the inputs so that inputs that have something in common are categorized together.
- The *statistical approach* to unsupervised learning is called ***density estimation***.

Note: pattern = example, instance, point, exemplar

3. Related fields of pattern recognition

- The methods used for pattern recognition have been developed in various fields, often independently.
- In *statistics*, going from particular observations to general descriptions is called ***inference***, learning (i.e., using training data) is called ***estimating*** and classification is ***discriminant analysis***.
- In *engineering*, classification is called *pattern recognition* and the approach is nonparametric and much more empirical.
- Other methods have their origins in *machine learning*, *artificial intelligence*, *artificial neural networks* and *data mining*.
- We will incorporate techniques from these different fields to give pattern recognition a more unified treatment.

Figure 1.2 Pattern recognition and related fields



4. Classification

- Classification is often the final step in a general process. It involves sorting objects into separate classes.
- In the case of an image, the acquired image is *segmented* to isolate different objects from each other and from the background, and the different objects are labeled.
- A typical pattern recognition system contains
 - a sensor,
 - a preprocessing mechanism (prior to segmentation),
 - a feature extraction mechanism,
 - a set of examples (training data) already classified and
 - a classification algorithm.

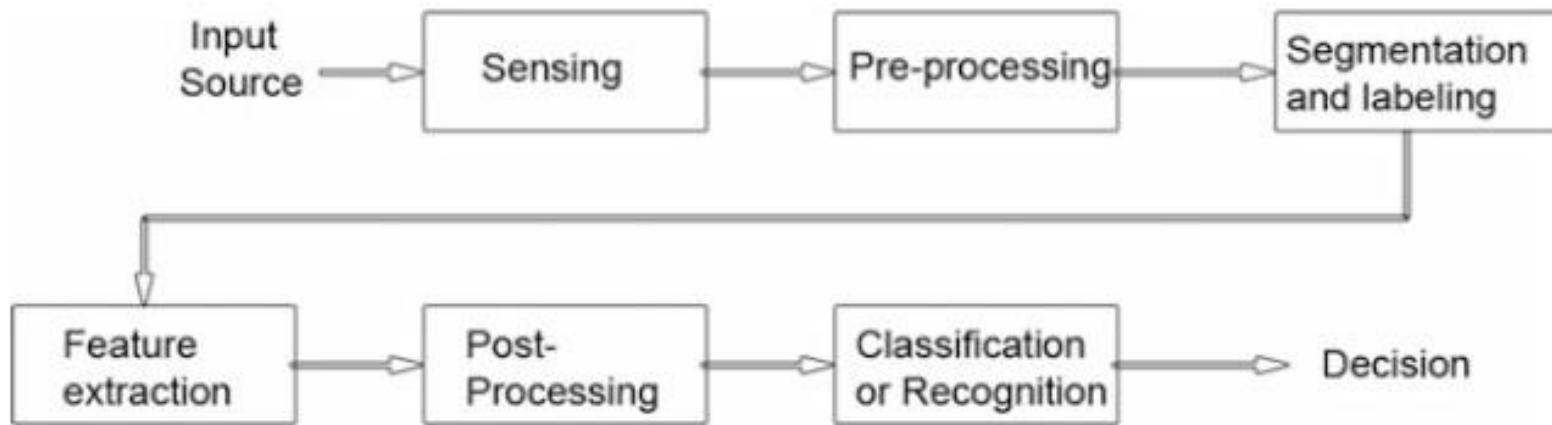


Figure 1.3 A general classification system

The ***feature extraction*** step reduces the data by measuring certain characteristic properties or features (such as size, shape, and *texture*) of the labeled objects.

These features (i.e. *values* of these features) are then passed to a ***classifier*** that evaluates the evidence presented and makes a decision regarding to class each object should be assigned, depending on whether a value of its feature fall inside or outside the tolerance of that class.

This process is used, e.g., in classifying lesions as benign or malignant.

- The quality of the acquired image depends on the resolution, sensitivity, bandwidth and signal-to-noise ratio of the imaging system. *Pre-processing* step such as image *enhancement* and image *restoration* may be required prior to segmentation, which is often a challenging process.
- The quality of the features is related to their ability to discriminate examples from different classes. Examples from the same class should have similar feature values, while examples from different classes should have different feature values, i.e. good features should have small **intra-class** variations and large **inter-class** variations (Figure 1.4).

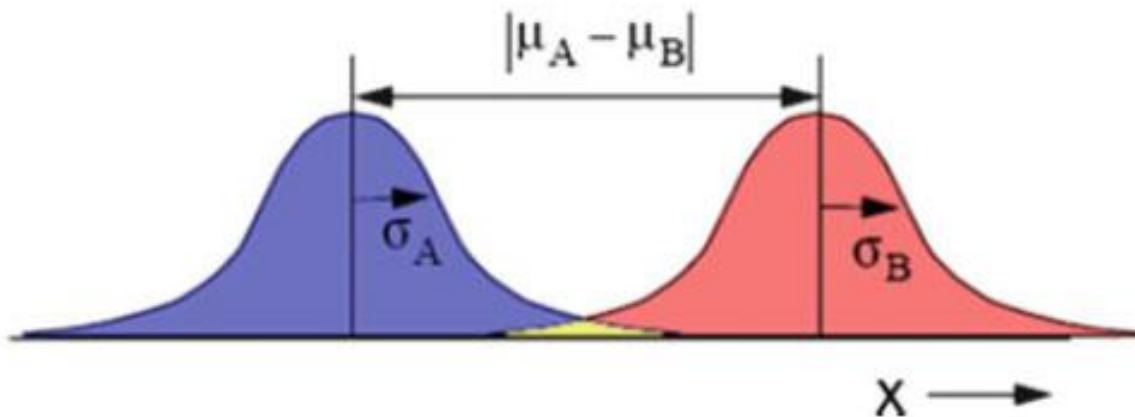


Figure 1.4 A good feature, x , measured for two different classes (*blue* and *red*) should have intra-class variations and large inter-class variations

- The features can be **continuous** or **categorical** or non-metric (i.e. qualitative). Categorical features can either be nominal (i.e. unordered) or ordinal.
- Humans are adept at recognize objects within an image, using size, shape, color, and other visual clues. They can do this despite the fact that the objects may appear from different *viewpoints* and under different *lightning conditions*, have different *sizes*, be *rotated* or when the images are partially *obstructed* from view.



Fig. 1.5 Face recognition needs to be able to handle different expressions, lighting and occlusions

- The goal of the classifier is to classify new data (***test data***) to one of the classes, characterized by a ***decision region***. The borders between decision regions are called ***decision boundaries***.

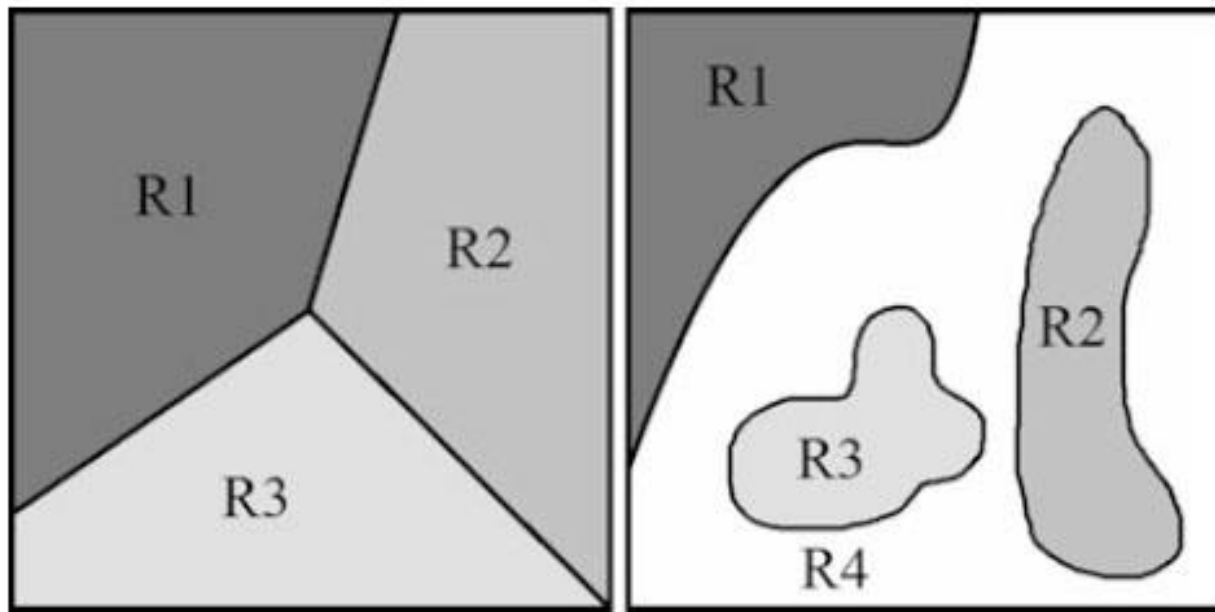


Figure 1.6 Classes mapped as decision regions, with decision boundaries

Datasets for pattern recognition

- There are a wide variety of datasets available on the Internet.
- One popular site is the machine learning repository at Univ. of California, Irvine:
<http://archive.ics.uci.edu/ml/datasets/>
- Large datasets used for data mining tasks are available at
kdd.ics.uci.edu
www.kdnuggets.com/datasets/

kdd: knowledge discovery and data mining

References

- G. Dougerty, 2013, *Pattern Recognition and Classification – An Introduction*, Springer
- S. Marshland, 2015, *Machine Learning – An Algorithmic Approach*, Chapman & Hall/CRC.
- M. N. Murty and V. S. Devi, 2011, *Pattern Recognition – An Algorithmic Approach*, Springer.

Terminology

- Machine learning: *học máy*, pattern: *mẫu*, pattern recognition: *nhận dạng mẫu*, feature: *đặc trưng*, feature extraction: *rút trích đặc trưng*, classification: *phân lớp*, discriminant analysis: *phân tích phân biệt*, training data: *dữ liệu huấn luyện*, test data: *dữ liệu thử*, image segmentation: *phân mảnh hình ảnh*, decision boundary: *biên quyết định*, decision region: *vùng quyết định*, pre-processing: *tiền xử lý*, continuous feature: *đặc trưng liên tục*, categorical feature: *đặc trưng rời rạc*, supervised learning: *học có giám sát*, unsupervised learning: *học không giám sát*, intra-class variation: *độ sai biệt trong nội bộ lớp*, inter-class variation: *độ sai biệt giữa các lớp*, sensor: *bộ cảm biến*, artificial neural network: *mạng nơ ron nhân tạo*, bioinformatics: *sinh tin học*, machine vision: *thị giác máy tính*, biometrics: *sinh trắc học*, remote sensing: *viễn thám*, density estimation: *ước lượng mật độ*, statistical pattern recognition: *nhận dạng mẫu thống kê*, syntactic pattern recognition: *nhận dạng mẫu ngữ pháp*.